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# 🌐 NIST's Role in Cloud Computing

- **Definition Authority**: NIST (National Institute of Standards and Technology) provides the widely accepted definition of cloud computing.
- **Framework & Standards**: It outlines essential characteristics, service models, and deployment models to ensure clarity and consistency across the industry.
- **Guidance**: Helps organizations adopt cloud securely and efficiently by publishing standards, guidelines, and best practices.
- **Neutral Reference**: Acts as a vendor-neutral body, ensuring definitions are not biased toward specific providers.

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# ⚡ Five Essential Characteristics of Cloud Computing (NIST)

1. **On-demand self-service**
   - Users can provision computing resources automatically without human interaction.
2. **Broad network access**
   - Services are available over the network and accessed through standard mechanisms (e.g., web, mobile).
3. **Resource pooling**
   - Provider's resources are pooled to serve multiple customers using a multi-tenant model.
4. **Rapid elasticity**
   - Resources can be scaled up or down quickly, appearing unlimited to users.
5. **Measured service**
   - Resource usage is monitored, controlled, and reported, enabling pay-per-use or metered billing.

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# 🏗 Deployment Models

| Model | Description | Example Use Case |
|---|---|---|
| **Public Cloud** | Services offered over the public internet, shared among multiple tenants. | AWS, Azure, Google Cloud |
| **Private Cloud** | Exclusive cloud infrastructure operated for a single organization. | On-premises VMware Cloud |
| **Hybrid Cloud** | Combination of public and private clouds with orchestration between them. | Sensitive data on private, scalable workloads on public |
| **Community Cloud** | Shared by several organizations with common concerns (e.g., compliance). | Government agencies sharing infrastructure |

# 🛠 Service Models

| Model | Description | Example |
|---|---|---|
| **IaaS (Infrastructure as a Service)** | Provides virtualized computing resources (servers, storage, networking). | Amazon EC2, Microsoft Azure VMs |
| **PaaS (Platform as a Service)** | Provides runtime environment, development tools, and frameworks. | Google App Engine, Heroku |
| **SaaS (Software as a Service)** | Delivers software applications over the internet. | Microsoft 365, Salesforce |

# 👥 Cloud Stakeholders

- **Cloud Provider**: Delivers cloud services (e.g., AWS, Azure).
- **Cloud Consumer**: Uses cloud services (individuals, enterprises).
- **Cloud Broker**: Manages use, performance, and delivery of cloud services across providers.
- **Cloud Auditor**: Provides independent assessment of cloud services (security, compliance).
- **Cloud Carrier**: Provides connectivity and transport of cloud services (ISPs, telecoms).